535510 RL Team Project: RL Reasoning for Recommendation Ranking

Pin-Yu Cheng, Nai-Wen Tsao, Chung-Kuang Chen
Department of Computer Science National
Yang Ming Chiao Tung University
{tony20040405.mg11, naiwentsao.mg11, conrade.cs14}@nycu.edu.tw

1 Project Overview

1.1 Profile

Track: Application track

Abstract and project goal:

This project proposes a novel framework that applies DeepSeek-R1-style Group Relative Policy Optimization (GRPO) to train large language models (LLMs) as reasoning-capable recommendation rankers.

The primary goal is to address the core limitations of existing LLM-based rankers, which typically rely on zero-shot prompting or supervised fine-tuning (SFT) and fail to align the model with listwise ranking objectives. Consequently, these conventional models suffer from a sharp degradation in ranking quality toward lower list positions, which is a direct result of sequence-level objectives being misaligned with metrics like NDCG. To overcome these challenges, our proposed method introduces rank-level reward decomposition coupled with a chain-of-thought reasoning trace. This mechanism effectively teaches the LLMs to reason before ranking, utilizing rank-level DCG rewards as clear, verifiable signals. Ultimately, this framework enables a verifiable, position-aware reinforcement learning (RL) training process that entirely eliminates the dependency on human annotations or proprietary models such as GPT-4o.

1.2 Motivation

• Why is the problem interesting?

Recommendation ranking is one of the most practically impactful applications of machine learning, but aligning LLMs with listwise ranking objectives remains fundamentally unsolved. Three observations motivate this work.

First, LLMs' ranking quality degrades sharply toward lower list positions — a consequence of sequence-level objectives misaligned with NDCG. Second, SFT-based rankers catastrophically forget general capabilities — GSM8K drops 28 pts, IFEval drops 27 pts after SFT [1][Lin et al., 2025]. This suggests that supervised learning by labeling might destroy LLMs' reasoning ability. Third, RLVR has proved reasoning can emerge from pure RL without human labels [2][DeepSeek-AI, 2025]. So here comes the question: can the same principle unlock better ranking?

• Critical challenges.

- Challenge 1: Non-causal credit assignment. Vanilla GRPO assigns one scalar reward per sequence. For ranked lists, only top-K positions drive NDCG, but back-prop spreads the gradient uniformly — ranks 1 and 5 receive identical updates.
- Challenge 2: Verifiable reward design. Unlike math tasks with exact answers, recommendation quality is relative to user preferences. Defining a clean, per-step verifiable reward without human annotation is non-trivial.
- Challenge 3: Catalog grounding. Open-vocabulary LLMs frequently hallucinate item IDs absent from the catalog. A warm-start pipeline that avoids dependency on proprietary models (e.g., GPT-4o) is needed.

• Justify why the problem remains open or unsolved.

- Non-causal credit assignment

  Rec-R1 [1][Lin et al., 2025] applies GRPO to recommendation but assigns a single sequence-level reward across all positions, causing non-causal credit assignment that disproportionately harms tail-position quality. More critically, Lin et al. also report that SFT-based warm-start leads to catastrophic forgetting of general capabilities, none of the RL-based recommendation works have explicitly addressed. ConvRec-R1 [3][Zhu et al., 2025], the current state-of-the-art, resolves the credit assignment problem through rank-level DCG decomposition (DCG@k:N) and demonstrates that small open-source LLMs can match GPT-4o on Recall@K and NDCG@K. However, it relies on GPT-4o to construct its SFT warm-start dataset, limiting reproducibility, and contains no explicit reasoning mechanism in its RL loop.

- Verifiable reward design

  On the reasoning side, DeepSeek-R1 [2] [DeepSeek-AI, 2025], building on the GRPO formulation of DeepSeekMath [4][Shao et al., 2024], demonstrates that chain-of-thought(CoT) reasoning can emerge from pure RL with verifiable rewards without human annotation. Moreover, it does not suffer the catastrophic forgetting observed in SFT-based approaches. However, this result has only been demonstrated in mathematical domains where rewards are binary and outputs are single answers, leaving open whether the same principle applies to ranking tasks with continuous, position-weighted rewards and catalog constraints.

- Catalog grounding

  LLMRank [5][Hou et al, 2024] demonstrates that zero-shot LLMs frequently coin item titles absent from the catalog and produce inconsistent formatting, directly reducing Catalog Hit Rate and rendering recommendations unusable in practice. This problem worsens in open generation settings where the model must produce a full ranked list without a provided candidate set.

• **State-of-the-art methods.** ConvRec-R1 [3][Zhu et al, 2025]. This work proposes a principled extension of GRPO tailored to rank-structured outputs, introducing rank-level advantage estimation based on causal reward decomposition (DCG@k:N) and a rank-level importance ratio computed via the geometric mean of item-token probabilities. On the REDDIT-V2 benchmark, a Llama-3.2-3B model trained with Rank-GRPO surpasses GPT-4o zero-shot on Recall@20 and NDCG@20, while a 0.5B model outperforms GPT-4o-mini — demonstrating that properly aligned small open-source LLMs can match or exceed much larger proprietary models in conversational recommendation. This establishes Rank-GRPO as the strongest published baseline and the direct foundation upon which our proposed framework builds.

2 Problem Formulation

We formulate the LLM-based recommendation ranking task as episodic Markov Decision Process (MDP), defined by the tuple $M = (S, A, R, \gamma, \pi_\phi)$. In this environment, the information available to the RL learner (the state $s \in S$) consists of the user's query, their sequential interaction history, and a top-K candidate list initially retrieved via methods like BM25 or a dense retriever. Based on this observed state, the agent (parameterized by an LLM policy π_ϕ) generates an action $a \in A$, which is a ranked list of N items. To facilitate reasoning during the generation process, this action can be optionally prefixed with a chain-of-thought trace enclosed within <think> tags.

The feedback signal provided to the learner is a composite scalar reward $R(s, a)$. The primary component is computed deterministically against held-out interactions using standard ranking metrics, specifically NDCG@K or Recall@K. To ensure structural compliance, an additional format reward $+\varepsilon$ is granted if the <think> block is present and the generated list parses correctly. The core optimization problem is to find a policy π_ϕ that maximizes the expected reward over the dataset, $\max_{\pi_\phi} E_{\{s\sim D,\, a\sim\pi_\phi\}}[R(s, a)]$, which is achieved via Group Relative Policy Optimization (GRPO) following an SFT warm-start. This formulation relies on three key assumptions: (1) the item catalog is finite and fully known at inference time; (2) ground-truth relevance labels exist for offline evaluation; and (3) the MDP is strictly episodic, meaning there are no cross-session dependencies between different user queries.

3 Empirical Evaluation

To comprehensively evaluate the effectiveness of our proposed method, we will conduct experiments focusing on ranking quality, computational efficiency, and output validity across established recommendation datasets.

3.1 Performance Metrics

We plan to adopt the following key performance metrics in our evaluation:

- NDCG@K: This will serve as our primary listwise ranking quality metric, as it is position-weighted and structurally aligns with our formulated reward signal.
- Recall@K: This metric will measure the proportion of relevant items successfully retrieved in the top-K, indicating the model's coverage of user-preferred items.
- Wall-Clock Time: We will track the training time per epoch on identical hardware to measure the computational overhead of our RL approach compared to the baselines.
- Catalog Hit Rate: We will track the percentage of generated ranked items that actually exist in the predefined catalog during evaluation. This is specifically required to monitor the hallucination rate of the LLM policy, ensuring the agent recommends real, valid items rather than generating fictitious titles.

3.2 Baseline Methods

We will compare our approach against the following baseline methods:

- LLMRank [5][Hou et al, 2024]: A zero-shot LLM ranker utilizing instruction prompting without any RL training, which establishes our fundamental prompting baseline.
- SFT-only (Ablation): This approach relies solely on supervised fine-tuning using behavioral cloning data to measure the specific performance gain achieved by adding RL on top of SFT.
- ConvRec-R1 [3][Zhu et al, 2025]: A state-of-the-art model utilizing rank-level GRPO (Rank-GRPO) with DCG position-weighted rewards and an SFT warm-start.

Method	R@5	R@10	R@15	R@20	N@5	N@10	N@15	N@20
Qwen-2.5-0.5B								
zero-shot	0.0022	0.0023	0.0023	0.0023	0.0025	0.0024	0.0023	0.0023
+SFT	0.0664	0.1020	0.1206	0.1316	0.0513	0.0629	0.0683	0.0712
+SFT+Rank-GRPO(exp_∞)	0.0936	0.1455	0.1796	0.2049	0.0742	0.0912	0.1012	0.1081

Method	R@5	R@10	R@15	R@20	N@5	N@10	N@15	N@20
Llama-3.2-3B-Instruct								
zero-shot	0.0523	0.0726	0.0817	0.0864	0.0463	0.0523	0.0548	0.0560
+SFT	0.0799	0.1211	0.1458	0.1604	0.0621	0.0753	0.0824	0.0863
+SFT+Rank-GRPO(exp_∞)	0.1169	0.1748	0.2134	0.2386	0.0916	0.1105	0.1218	0.1287

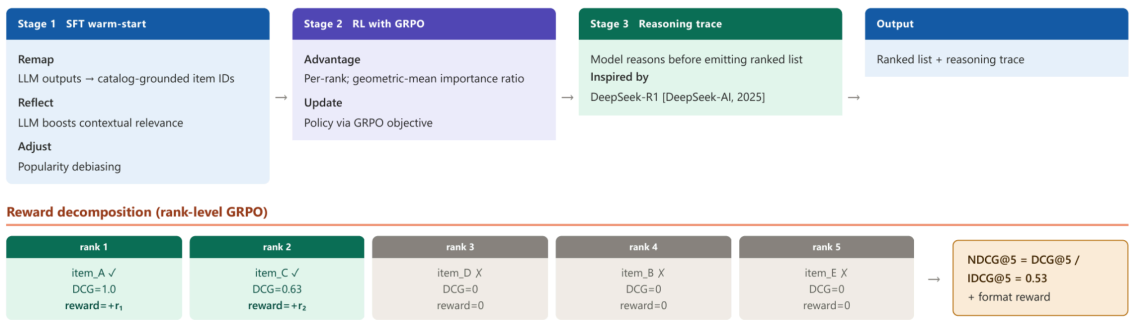
SFT transfers catalog grounding and output formatting, but plateaus there because next-token cross-entropy does not directly optimize NDCG and inherits the teacher's tail-position weakness. Adding Rank-GRPO further improves NDCG@20 by +52% relative for Qwen and +49% for Llama-3B, with the relative gain growing monotonically in K (Qwen: +45/45/48/52% at K = 5/10/15/20) — the diagnostic signature of rank-level credit assignment, since sequence-level objectives would dilute the gradient uniformly.

3.3 Benchmark Tasks or Datasets

We plan to evaluate our method using the following standard datasets:

- MovieLens-1M: A dataset containing approximately 1 million interactions, serving as a standard sequential recommendation benchmark for LLM4Rec tasks [5][Hou et al, 2024].
- Reddit-v2: A dataset comprising roughly 400K conversations, which will function as our primary conversational recommendation benchmark [3][Zhu et al, 2025].

4 Methodology



5 Experimental Results

5.1 Evaluation of Baseline Methods

5.2 Evaluation of toy task

5.2.1 Rank-level and sequence-level reward

Toy task settings:

Genres: 8 (sci-fi, romance, horror, comedy, crime, animation, drama, fantasy)

Year range: 1980-2025

Feature representation (5-dim): [0] genre_overlap: len(user_genres ∩ item_genres) / 2.0 [1] popularity: item.popularity ∈ [0, 1] [2] recency: (item.year - 1980) / 45 [3] dislike_penalty: -1.0 if disliked_genre in item_genres else 0.0 [4] bias: 1.0

Weight vector: 5-dim parameter Initial weights: [0.01, 0.85, 0.50, 0.0, 0.0] → heavily favor popularity (0.85) and recency (0.50) → almost ignore genre matching (0.01)

Evaluation: Metric: NDCG@5, Recall@5 Ranking function: greedy (argmax scores)

Initial biased policy (weights [0.01, 0.85, 0.50]) produces ranking:

Rank 1: movie_0100 (high popularity) → rel=0

Rank 2: movie_0042 (good popularity) → rel=1 ← only relevant item at position 2

Rank 3: movie_0015 (high recency) → rel=0

Rank 4: movie_0187 (moderate popularity) → rel=1 ← another relevant at position 4

Rank 5: movie_0099 (high popularity) → rel=0

Relevance vector: [0, 1, 0, 1, 0] Sequence reward: NDCG@5 = $(0 + 1/\log_2(3) + 0 + 1/\log_2(5))$ / IDCG = (0.631 + 0.4307) / 2.131 = 0.5

sequence-level: scalar advantage shared across all positions

rank-level: position-specific DCG contribution normalized by IDCG

- rank-level reward receives position-specific advantage:
- advantage[t] = (rank_reward[t] - μ) / σ , loss = -Σ_t advantage[t] * log π(a_t) where rank_reward[t] = rel[t] * discount[t] / IDCG
- Training progress:
  - Step 10: NDCG@5 = 0.8190, Recall@5 = 0.6900
  - Step 20: NDCG@5 = 0.9925, Recall@5 = 0.8746
  - Step 30: NDCG@5 = 1.0000, Recall@5 = 0.8831
  - Improvement: NDCG@5 +0.6928
- sequence-level reward receives one advantage signal per rollout:
- advantage = (seq_reward - μ) / σ, loss = -advantage * Σ_t log π(a_t)
- Training progress:
  - Step 10: NDCG@5 = 0.9266, Recall@5 = 0.7965
  - Step 20: NDCG@5 = 1.0000, Recall@5 = 0.8831
  - Step 30: NDCG@5 = 1.0000, Recall@5 = 0.8831
  - Improvement: NDCG@5 +0.6928

5.2.2 Reasoning format + Catalog grounding reward

Toy task settings:

- Items Catalog = { (Interstellar, 2014), (Arrival, 2016), (The Matrix, 1999), (Toy Story, 1995), (The Godfather, 1972), (Spirited Away, 2001)}

- Recommendation Groundtruth = {(Arrival, 2016), (Interstellar, 2014)}

- Seen Titles = {(The Matrix)}, this indicates the user has already interacted with The Matrix. Therefore, recommending (The Matrix, 1999) again should be avoided, even if it exists in the catalog.

- format_reward : 1 if the response contains both complete <think>...</think> and <answer>...</answer> sections; otherwise 0. It evaluates whether the recommender's response strictly follows the required output format

- catalog_hit_rate : The proportion of raw recommended items that exist in the catalog, used to measure LLM hallucination of non-existent items.

Total reward = NDCG@5 + 0.05 * format_reward + 0.05 * catalog_hit_rate

Check the format of <think>...</think><answer>...</answer>

↓

Analysis of the Movie Title (YYYY) in <answer>

↓

Calculate catalog_hit_rate

↓

Remove seen list items and items not in the catalog

↓

Caculate NDCG@5

↓

Combine Total Reward and update the model

Here are four examples of this task.

- valid_reasoning_good_rank

```
"""<think>The user likes cerebral science fiction, so I should rank grounded sci-fi films first.</think>
<answer>
1. Arrival (2016)
2. Interstellar (2014)
3. Spirited Away (2001)
4. Toy Story (1995)
5. The Godfather (1972)
</answer>"""
```

Total reward = 1 + 0.05 * 1 + 0.05 * 1 = 1.1

- no_think_but_good_rank

```
"""<answer>
1. Arrival (2016)
2. Interstellar (2014)
3. Spirited Away (2001)
</answer>"""
```

Total reward = 1 + 0.05 * 0 (no reasoning) + 0.05 * 1 = 1.05

- hallucinated_items

```
"""<think>I will recommend famous sci-fi.</think>
<answer>
1. Arrival (2016)
2. Fake Space Movie (2020)
3. Another Unknown Film (2018)
</answer>"""
```

(Arrival, 2016) is at 1st, and the answer should hit 1st and 2nd .

NDCG = DCG / IDCG = $(1/\log_2(1+1)) / (1/\log_2(1+1) + 1/\log_2(2+1)) = 0.6131$

Total reward = 0.6131 + 0.05 * 1 + 0.05 * ⅓ (only Arrival is in the catalog) = 0.6798

- bad_rank

```
"""<think>I ignored the preference.</think>
<answer>
1. Toy Story (1995)
2. The Godfather (1972)
3. Arrival (2016)
4. Interstellar (2014)
</answer>"""
```

(Arrival, 2016) is at 3rd and (Interstellar, 2014) is at 4th.

NDCG = $(1/\log_2(3+1) + 1/\log_2(4+1)) / (1/\log_2(1+1) + 1/\log_2(2+1)) = 0.5706$

Total reward = 0.5706 + 0.05 * 1 + 0.05 * 1 = 0.6706

In the reward settings, hallucinated_items (0.6798) is slightly higher than bad_rank (0.6706). Top-rank relevance takes precedence over catalog completeness. If you place more emphasis on catalog grounding, you can increase catalog weight or add hallucination penalty.

6 Expected Contributions of Each Team Member

Please describe how you would collaborate with each other and specifically what each member would contribute to this project.

- Pin-Yu Cheng: toy example dataset finding and literature review
- Nai-Wen Tsao: problem formulation and empirical evaluation.
- Chung-Kuang Chen: problem definition and small LLM research

As you just kick off the project, it is very likely that you may need to reallocate the tasks among yourselves as you proceed. Please make sure that you reach a consensus on the expected contributions.

References

[1] Lin, Jiacheng, Tian Wang, and Kun Qian. "Rec-r1: Bridging generative large language models and user-centric recommendation systems via reinforcement learning." arXiv preprint arXiv:2503.24289 (2025).

[2] Guo, Daya, et al. "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning." arXiv preprint arXiv:2501.12948 (2025).

[3] Zhu, Yaochen, et al. "Rank-GRPO: Training LLM-based Conversational Recommender Systems with Reinforcement Learning." arXiv preprint arXiv:2510.20150 (2025).

[4] Shao, Zhihong, et al. "Deepseekmath: Pushing the limits of mathematical reasoning in open language models." arXiv preprint arXiv:2402.03300 (2024).

[5] Hou, Yupeng, et al. "Large language models are zero-shot rankers for recommender systems." European conference on information retrieval. Cham: Springer Nature Switzerland, 2024.